Timing Search†

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Abstract

Why do different industries fluctuate at different frequencies? Because each industry has its own profit curvature (φ) and timing friction (μ), and these two primitives—not the shock process—set the cycle length. I develop a *timing-search* model: agents choose when to enter a market under frictions that penalize rapid adjustment. The indifference condition across entry dates prevents the industry from settling at its optimum—entry must persist past optimal scale, forcing a return—and cycles arise at frequency $\omega = \sqrt{\varphi/\mu}$. Cycle frequency and the per-firm welfare cost of cycles have disjoint primitives: market structure (φ, μ) sets frequency; shock variance and bottleneck dissipation set welfare. Empirically, the predicted slope is recovered across 16 manufacturing industries with historical concentration as instrument; entry inputs flow flexibly across industries while the profit landscape does not, suggesting the profit landscape as the operative cross-industry channel. Entry-exit oscillations, post-shock periodicity, and a post-COVID overshoot are additional consistent diagnostics.

Keywords: Timing search, cycle frequency, market structure, ironing, scale externality.

JEL codes: D83, E32, L13, R12.

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1 Introduction

U.S. industries fluctuate at noticeably different frequencies: dominant spectral periods range from 2.6 to 6.6 years in industrial production and from 2.4 to nearly 10 years in establishment entry. Why do different industries fluctuate at different frequencies? Standard business-cycle frameworks derive cycle properties from a common shock process (Kydland and Prescott, 1982) or policy rule (Christiano et al., 2005), leaving cross-industry frequency variation unexplained. Two obstacles stand in the way of a structural account. First, in models where cycle frequency is inherited from shock persistence or policy, industries hit by common shocks should share a common cycle length—a prediction the data reject. Second, existing endogenous-cycle models generate internal persistence but do not tie cycle frequency to industry-specific market-structure primitives, nor separate the determinants of timing from those of welfare. This paper addresses both by introducing *timing search*: agents choose *when* to enter under matching frictions that prevent instantaneous adjustment. Indifference across entry dates in equilibrium leads to oscillation rather than monotone return to optimal scale, with cycle frequency taking the closed form

$$\omega \;=\; \sqrt{\varphi/\mu}, \tag{1}$$

where φ is the curvature of profits around optimal industry scale and μ is the entry-matching friction—both market-structure primitives, not properties of the shock process.

The analysis delivers three results. First, I propose a mechanism for empirically-supported industry-specific cycles. Under timing search—agents choosing *when* to enter under matching frictions that penalize rapid adjustment—indifference across entry dates rules out settling at the optimum and reverses the flow once the industry overshoots, generating a cycle with closed-form frequency $\omega = \sqrt{\varphi/\mu}$ (Theorem 1, Section 3). A distinctive prediction is that the cycle period depends on market structure alone, not on cycle amplitude—so deeper recessions recover at proportionally faster speeds in the same time as shallow ones. This matches the plucking pattern of Friedman (1993), confirmed in U.S. industrial production over 1972–2024 (Pearson $r = +0.80$, $p = 0.029$ across the seven NBER recessions in the sample; Section 5.4).

Second, I bring the prediction that cycle frequency is pinned by market structure rather than by shocks to U.S. industry data (Section 5). The headline test regresses the spectral peak frequency on the Ellison and Glaeser (1997) geographic concentration index γ_i across 16 manufacturing 3-digit industries. The OLS slope is $\hat{\beta}_\gamma = 0.20$ ($p = 0.010$, $R^2 = 0.39$); instrumenting current γ_i with the 1947 Hoover localization coefficient from Kim (1995) raises the estimate to $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.45$ (HC1 SE 0.17), statistically indistinguishable from the structural prediction $\beta_\gamma = 1/2$ under conventional Wald (1943) inference and Anderson and Rubin (1949) weak-IV-robust inference (first-stage $F = 5.30$). The same sign holds in a parallel BDS subsector cross-section using independent data on both cycle frequency and concentration.

This evidence is consistent with—but does not directly identify—a profit-curvature channel: across manufacturing, entry inputs (permits, construction crews, equipment supply chains) flow relatively freely, so the matching friction varies little across 3-digit industries, and the cross-industry signal in cycle frequency comes from differences in the profit landscape. Three additional diagnostics are consistent with the mechanism but rest on weaker identification: BDS net-entry rates show oscillatory features outside the AR(1) class in 10 of 19 sectors, manufacturing and oil investment oscillate at industry-specific frequencies after large historical shocks ($T = 5.0$ years post-WWII; $T = 3.25$ years post-1979), and 14 of 19 BDS sectors overshoot above their pre-pandemic trend after COVID-19. Standard benchmarks (AR(1) RBC, NK AR(2), Beaudry et al. (2020), and a multi-sector RBC with input-output linkages) capture only subsets of the joint pattern; reproducing it at observed magnitudes requires a mechanism that maps industry-level primitives into industry-level frequencies (Online Appendix F).

Third, I show which entry-side policies move the welfare cost of cycles and which do not, delivering a sharp policy taxonomy (Theorem 2, Section 6.2). The taxonomy follows from a frequency-welfare separation that the model delivers rather than imposes: $\omega_0 = \sqrt{\varphi/\mu}$ depends only on the market-structure pair (φ, μ), while the welfare loss $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ depends only on the disjoint pair of profit-shock variance and entry-pipeline friction χ. Two natural classes of policy therefore fall on the timing-only side: streamlining entry matching (e.g., faster permitting) and strengthening agglomeration (e.g., cluster subsidies) change cycle speed without affecting welfare cost. Only policies that reduce entry-pipeline congestion or stabilize profit shocks move welfare. The taxonomy thus distinguishes welfare-improving policy from policy that, while visibly altering cycle dynamics, leaves welfare cost unchanged.

Related literature This paper builds on four literatures: structural and endogenous cycles, search and matching frictions, industry dynamics, and agglomeration.

Structural and endogenous cycles. The structural-cycle tradition is anchored by Friedman (1993), who reads recessions as plucks below an unobserved natural ceiling rather than as symmetric stochastic deviations; the deeper-recession-faster-recovery pattern that view predicts is revisited empirically by Bordo and Haubrich (2017) and Dupraz et al. (2019), and emerges in the present model as a corollary of isochrony (Section 3.5). The closest formal macroeconomic antecedent is Beaudry et al. (2020), who show that financial frictions can generate endogenous boom-bust dynamics through a stochastic limit cycle (extended in Beaudry et al. (2024)); the present paper builds on the broader insight that short-lived shocks can produce persistent cyclical adjustment, but addresses a different question—why do different industries exhibit different characteristic frequencies?—and answers it through timing search and market-structure primitives rather than through a financial-friction parameterization. The animal-spirits and self-fulfilling-prophecies tradition of Farmer and Guo (1994) and Farmer (1999) generates endogenous fluctuations from indeterminacy and belief shocks; here, by contrast, fluctuations arise from determinate primitives without indeterminacy. Beaudry and Portier (2006) document stock prices leading TFP; Grandmont (1985), Azariadis (1981), Benhabib and Farmer (1994), and Goodwin (1967) provide earlier endogenous-cycle precedents. The closest timing-based antecedent is Jovanovic (2009), who shows how investment timing can generate endogenous business cycles; the present paper shifts the object from the option value of investment timing to the equilibrium distribution of entry dates, which yields a closed-form mapping from market-structure primitives to cycle frequency.

Search and matching frictions. Timing search is in the spirit of Stigler (1961), McCall (1970), and especially Burdett and Mortensen (1998): a temporal counterpart in which agents choose among dates rather than among contemporaneous alternatives, with an ironing condition analogous to the equal-value condition in Burdett and Mortensen (1998) (Online Appendix A.1). The micro-foundation $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ further connects to the Diamond-Mortensen-Pissarides literature (Mortensen and Pissarides, 1994; Diamond, 1982; Pissarides, 2000; Petrongolo and Pissarides, 2001; Shimer, 2005): matching frictions in entry generate equilibrium business cycles in the same way DMP matching frictions generate equilibrium unemployment.

Industry dynamics. The model isolates a timing-search channel within a deliberately stripped-down entry environment, complementary to the richer heterogeneity and selection mechanisms of Hopenhayn (1992) and Hopenhayn and Rogerson (1993); the entry-wave dynamics generated by the ironing condition complement Jovanovic (1982)'s learning-based exit explanation, and Davis and Haltiwanger (1992) document the entry/exit empirics this literature studies. A separate strand on aggregate fluctuations from disaggregated origins (Gabaix, 2011; Acemoglu et al., 2012) provides the natural alternative the paper tests against in Online Appendix F.2.

Agglomeration. The scale-congestion logic of Krugman (1991), Duranton and Puga (2004), Rosenthal and Strange (2004), Ellison and Glaeser (1997), and Ellison et al. (2010) is transposed from space to time: combined with timing frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters), and the EG concentration index serves as an external proxy for the sharpness of the scale-congestion trade-off in the present framework; Syverson (2004) provides direct evidence of the productivity-density relationship that underlies the hump-shaped profit assumption.

2 Environment

The environment has three elements chosen to make the cycle-frequency result precise: a hump-shaped profit landscape that gives the industry a well-defined optimal scale, a matching function for entry that endogenously generates a quadratic adjustment cost, and linear-utility households that let industries evolve independently. Section 2.1 sets up the agents, Section 2.2 the profit landscape, Section 2.3 the timing friction, and Section 2.4 combines them into the reduced-form dynamics that drive the equilibrium.

2.1 Agents

The economy consists of two types of agents: a representative household that consumes and supplies labor, and a fixed mass of potential entrepreneurs who alternate between operating firms and searching for entry opportunities. The household has linear utility over consumption, $U = \int_0^\infty e^{-\rho t} C(t)\, \mathrm{d}t$, and supplies labor perfectly elastically at a fixed wage $\bar{w}$.[1] Linear utility is not merely a simplification: it eliminates wealth

[1]Equivalently, the household has linear utility in consumption and linear disutility from labor, $U = \int_0^\infty e^{-\rho t} \big[C(t) - \bar{w}\, L(t)\big]\, \mathrm{d}t$, so the static labor-supply condition pins $w = \bar{w}$ and labor is supplied elastically. This is in the spirit of Greenwood et al. (1988) preferences in eliminating the wealth effect

effects and cross-industry price feedback, making the single-industry analysis a general equilibrium result (Section 3, after Theorem 1). Three ingredients define the entry side of the environment.

Assumption 1 (Entry economy)**.**
A mass $M > 0$ of potential entrepreneurs, each at any instant either active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor hired from the representative household at wage $\bar{w}$, with technology $y_i = zl_i^\gamma$, $\gamma \in (0, 1)$. Each entry incurs a per-firm sunk cost $\kappa \geq 0$, paid at the moment of entry. Firms exit at Poisson rate $\delta > 0$, returning to the search pool.*

Let $n(t)$ denote the mass of active production units at date t and $S(t) = M - n(t)$ the mass of searchers. Throughout the paper, I interpret n as the mass of active *establishments*—physical production units such as plants, stores, or branches—rather than firms (legal entities). The establishment interpretation matches both empirical series used in Sections 5–5.4: BEA investment in structures measures the rate of new establishment construction, and BDS establishment entry and exit rates measure establishment births and deaths. The matching-congestion microfoundation of μ (Section 2.3) is also most natural at the establishment level, where bottlenecks in permits, specialist labor, equipment, and sites constrain rapid adjustment. The aggregate state of the economy is fully characterized by $n(t)$ and its rate of change $\dot{n}(t)$.

2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = zl_i^\gamma \cdot m(n)$. Profit is $\pi(n) = \bar{\pi}(w) \cdot m(n)$, where $\bar{\pi}$ is the standard profit given the wage.[2]

on labor supply.

[2]Solving the firm's static problem $\max_l zl^\gamma - wl$ gives optimal labor $l^* = (\gamma z/w)^{1/(1-\gamma)}$ and standard profit $\bar{\pi}(w) = (1-\gamma)\gamma^{\gamma/(1-\gamma)}z^{1/(1-\gamma)}w^{-\gamma/(1-\gamma)} > 0$. With $m(n_p) = 1$ in the Gaussian specification (Assumption 2), the constant π_0 in equation (3) equals $\bar{\pi}(\bar{w})$.

Assumption 2 (Gaussian scale efficiency)**.**
Scale efficiency takes the Gaussian form:

$$m(n) = \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

Interpretation of Assumption 2. The function peaks at $n = n_p$ (optimal market scale) and falls off symmetrically. The parameter φ governs the sharpness of the trade-off: high φ means a narrow peak (sharp trade-off); low φ means a broad peak (robust to deviations). The hump-shaped specification is consistent with the productivity-density relationship documented in Syverson (2004), in which establishment-level productivity rises and then falls with market-level density—the empirical pattern this paper's profit landscape encodes. Online Appendix A.2 provides a matching-theoretic microfoundation.

For the analytical results, I work with the reduced-form Gaussian profit:

$$\pi(n) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \pi_0 > 0. \tag{3}$$

The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity delivers exact harmonic oscillations (Theorem 1). The general case—where wage congestion adds a non-quadratic correction to $\ln \pi$—produces anharmonic oscillations with the same leading-order frequency (Online Appendix B.2).

2.3 The timing friction: congestion in the entry process

Entry is mediated by a Cobb-Douglas matching technology $m(S, R) = S^\alpha R^{1-\alpha}$ with elasticity $\alpha \in (0, 1)$, where the $S(t) = M - n(t)$ searchers compete for R units of *entry resources*—permits, specialized labor, venture capital, factory sites—to form matches. The resource stock R is endogenously supplied by an infrastructure sector; the matching-friction derivation in this subsection uses R at its steady-state level R_{ss} to pin the timing friction μ. Deviations of R from R_{ss}—driven by net adjustment pressure $\dot{n}$ in the stationary cycle—generate an additional dissipation channel that pins the cycle amplitude, taken up in Section 4.1. The matching infrastructure is tuned to clear the steady-state throughput $e_{ss} = \delta n_p$, with routine replacement at rate $\delta n(t)$ serviced smoothly. The timing friction operates on net adjustment pressure $\dot{n}$—deviations of the adjustment-speed margin from zero: rapid entry ($\dot{n} > 0$) overloads permitting and construction, while rapid net contraction ($\dot{n} < 0$) overloads resale and reorganization. Both reduce the matching probability per searcher and produce the timing friction.

Assumption 3 (Timing friction)**.**
The multiplicative timing friction takes the log-quadratic form

$$\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}, \qquad \mu > 0,$$

where μ is the friction parameter.

The functional form arises as the continuous-time limit of *proportional adjustment costs* in the entry pipeline: in a discrete-time economy where effective output is reduced by a fraction proportional to the squared adjustment rate $(\Delta n/\Delta)^2$ each period, taking the period length $\Delta \to 0$ yields the exponential-quadratic form, with μ a free friction parameter (Online Appendix A.3). The exponential-quadratic specification is canonical: any symmetric friction ξ with $\xi(0) = 1$ and $\xi'(0) = 0$ admits the second-order expansion $\ln \xi(\dot{n}) = -\frac{\mu}{2}\dot{n}^2 + O(\dot{n}^4)$, and the leading-order frequency $\omega = \sqrt{\varphi/\mu}$ depends only on the second-order coefficient. Symmetry of the flow penalty in $\dot{n}$ is itself a knife-edge condition: under asymmetric strain, the leading-order ironing condition would carry a cubic term in $\dot{n}$ that produces amplitude-dependent frequency (Online Appendix B.4); the exact-harmonic result of Theorem 1 requires the symmetric specification, while the qualitative cycle survives asymmetric strain.

Combining Assumption 3 with the Cobb-Douglas matching technology described above pins the friction parameter to a specific function of structural primitives: any symmetric strain term $g(\dot{n}^2/(\delta n_p)^2)$ with $g(0) = 0$ and $g'(0) = 1/2$ multiplying the matching probability gives, at second order around $\dot{n} = 0$,

$$\mu = \frac{1-\alpha}{\alpha\,(\delta n_p)^2}, \tag{5}$$

independent of the higher-order shape of g (Online Appendix A.4). Here α is the matching elasticity and δn_p is the steady-state replacement throughput. Equation (5) is the calibration we adopt when reading μ off the matching microfoundation; the body's results that depend on μ alone (e.g., $\omega = \sqrt{\varphi/\mu}$) hold under Assumption 3 for any positive friction parameter.

The timing friction μ decreases in the matching elasticity α and decreases in the steady-state throughput δn_p (the entry-resource stock R is absorbed into the steady-state calibration of n_p via the matching technology and therefore does not appear in (5) as a separate factor). Substituting (5) into $\omega = \sqrt{\varphi/\mu}$ gives the structural frequency

$$\omega = \delta n_p \sqrt{\alpha\varphi/(1-\alpha)}, \tag{6}$$

connecting cycle frequency to three independently measurable objects: matching elasticity α, profit curvature φ, and steady-state turnover δn_p. In the absence of direct estimates of entry-matching elasticities, we use the labor-market range $[0.5, 0.7]$ from Petrongolo and Pissarides (2001) as a benchmark; the structural prediction $\omega = \sqrt{\varphi/\mu}$ is robust to $\alpha \in [0.3, 0.8]$ since μ varies smoothly with α over this range. Effective output is $y(t) = \pi(n(t))\,\xi(\dot{n}(t))$. The quadratic form in $\dot{n}$ arises generically from any smooth symmetric matching friction; Online Appendix A.3 derives the same ξ from discrete-time proportional adjustment costs.

2.4 Combined structure

Combining the Gaussian profit and the Gaussian friction, effective output is:

$$y(t) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\dot{n}^2\Big). \tag{7}$$

The effective output is a bivariate Gaussian kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: n_p for market scale and $\dot{n} = 0$ for adjustment speed.

3 Ironing Along Time: Equilibrium and the Main Result

The equilibrium condition that agents cannot profit by retiming their entry—the ironing condition—pins a rate of entry along any approach to optimal scale. That pinned rate becomes momentum the industry cannot discard at the optimum without creating arbitrage. The result is cycles, and in the frictionless limit they are exactly harmonic with frequency $\omega = \sqrt{\varphi/\mu}$. This section formalizes the argument.

3.1 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning effective flow payoff $\pi(n(t))\xi(\dot{n}(t))$ per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, \pi(n(t))\, \xi(\dot{n}(t))\, \mathrm{d}t. \tag{8}$$

$V(\tau)$ is the value of a single entry spell—operating from τ until stochastic exit. Re-entry after exogenous exit is treated as a fresh draw of the same problem for whichever agent occupies the search pool at that date, so the single-spell formulation is without loss for equilibrium dynamics under the ironing condition. The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n(t)\}$ as given. An agent who does not enter at date t remains in the search pool, with continuation value

$$V^{\text{search}}(t) = \max_{\tau \geq t} e^{-r(\tau-t)}\big[V(\tau) - \kappa\big], \tag{9}$$

where κ is the per-firm sunk entry cost from Assumption 1, paid at the moment of entry. A second cost-side parameter, the bottleneck friction χ, summarizes the convex resource cost the supplier sector pays to deliver net adjustment $\dot{n}$ at peak strain (Section 4.1). The two are structurally distinct: κ pins the steady-state ironing level but does not dissipate, while χ produces the damping force $-\chi\dot{n}$ that pins cycle amplitude and enters the welfare formula (Section 6.1).

3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date. Together with the initial condition $n(0)$, it determines the entire path of the economy through the aggregate consistency condition $\dot{n} = e - \delta n$.

Definition 1 (Timing search equilibrium)**.**
A timing search equilibrium *is a path $\{n(t)\}_{t\geq 0}$ and an entry policy $e(\cdot)$ such that:*

(i) ***Individual optimality****: Each agent chooses her entry date to maximize $V(\tau)$.*

(ii) ***Temporal free entry (with complementary slackness)***: $V(t) \le V^{search}(t) + \kappa$ *at all dates, with equality whenever* $e(t) > 0$ *and* $n(t) < M$. *At dates where the population cap binds* ($n(t) = M$) *or the entry flow is zero* ($e(t) = 0$)*, the inequality may be strict.*

(iii) ***No profitable timing deviation***: $V(\tau) \le V(t)$ *for all* τ *and all* t *in the support of entry.*

(iv) ***Aggregate consistency***: $\dot{n}(t) = e(t) - \delta\, n(t)$, *where* $e(t) \ge 0$.

(v) ***Feasibility***: $n(t) \le M$ *for all* t.

The no-profitable-timing-deviation condition Condition (iii) is the key requirement. In Burdett and Mortensen (1998), the analogous condition is that no firm can profit by reposting its wage, which yields equal profits across wages. Here, the condition that no agent can profit by retiming her entry yields equal profits across dates. If any date offered strictly higher value, agents would crowd that date, changing $n(t)$ until the advantage was eliminated.

Commitment versus dynamic optimization The formulation in (8)–(9) is a commitment problem: agents choose τ taking the entire future path $\{n(t)\}$ as given. An alternative formulation would have agents observe the current state $(n(t), \dot{n}(t))$ at each instant and decide dynamically whether to enter. The ironing condition makes the two equivalent. Since $V(\tau) = V^*$ for all τ in equilibrium, an agent who could re-optimize at any date t would find no profitable deviation—she is indifferent across all entry dates. The commitment equilibrium is therefore also an equilibrium of the dynamic problem: agents who can freely re-time their entry choose not to.

3.3 The ironing condition

The key equilibrium condition is that agents cannot profit by retiming their entry. This is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998): Burdett-Mortensen equalize profits across wages so that no firm gains by reposting, and timing search equalizes payoffs across dates so that no agent gains by retiming. I call this the *ironing condition*, since it “irons out” any temporal variation in the value of entry.

Proposition 1 (Ironing condition)**.**
In a timing search equilibrium where entry occurs at all dates, agents are indifferent across entry dates. Equivalently, the effective flow payoff is equalized:

$$\pi(n(t))\,\xi(\dot{n}(t)) = C \qquad \textit{for all } t, \tag{10}$$

for some constant $C > 0$.

Proof. Suppose entry occurs at all dates. For any two dates τ_1, τ_2 in the support of entry, condition (iii) of Definition 1 requires $V(\tau_1) \leq V(\tau_2)$ and $V(\tau_2) \leq V(\tau_1)$, so $V(\tau_1) = V(\tau_2)$. Since the support is all of $\mathbb{R}_+$, $V(\tau) = V^*$ for all $\tau \geq 0$. The equilibrium path $n(\cdot)$ is C^1 by aggregate consistency (condition (iv)) and the smoothness of π and ξ, so $V(\tau)$ is differentiable in τ.[3] Differentiating (8):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since V is constant): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)V^* \equiv C$. ■

Regularity condition The condition that entry occurs at all dates is satisfied whenever the searcher pool $S(t) = M - n(t)$ is bounded away from zero along the equilibrium path and the timing friction μ is positive. With a large population M and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix B.3 provides a formal verification.

3.4 The main result

The equilibrium under Gaussian profits and zero bottleneck friction is an exact harmonic oscillator with frequency $\omega = \sqrt{\varphi/\mu}$ (Theorem 1). This subsection derives that result. The frictionless case is not the most realistic specification, but it is the analytically tractable one: the ironing condition reduces to a frictionless harmonic oscillator whose orbit and period admit closed-form expressions at every amplitude.

Section 4 extends this frictionless oscillator with bottleneck friction and stochastic productivity shocks; the same closed-form natural frequency $\omega = \sqrt{\varphi/\mu}$ carries over

[3] The differentiability of V in τ requires that $t \mapsto \pi(n(t))\xi(\dot{n}(t))$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of π (Assumption 2) and ξ (Section 2.3), combined with the continuity of the entry flow $e(\cdot)$ in condition (iv).

to the stationary cycle of that section, where it locates the spectral peak of $n(t)$ up to a small bottleneck-damping correction (Proposition 3).

Taking logarithms of the ironing condition (10), substituting the functional forms (7), and rearranging:

$$\underbrace{\frac{\varphi}{2}(n-n_p)^2}_{\text{profit loss}} + \underbrace{\frac{\mu}{2}\,\dot{n}^2}_{\text{adjustment cost}} = \underbrace{E}_{\substack{\text{equilibrium entry}\\ \text{value (constant)}}}, \tag{11}$$

where $E = \ln \pi_0 - \ln C > 0$ is the equilibrium value level pinned by the ironing condition (equal value across all entry dates). Differentiating Equation (11) with respect to time gives $\dot{n}\,[\varphi(n-n_p) + \mu\ddot{n}] = 0$; dividing through by $\dot{n}$, which is nonzero along the orbit, yields the corresponding law of motion:

$$\underbrace{\ddot{n}}_{\substack{\text{change in}\\ \text{entry rate}}} = \underbrace{-\frac{\varphi}{\mu}\,(n-n_p)}_{\substack{\text{pull toward}\\ \text{optimal scale}}}. \tag{12}$$

At every instant the change in the entry rate is pinned by the current deviation from optimal scale, scaled by the ratio of profit curvature to timing friction φ/μ: industries far from n_p accelerate quickly back toward it, industries near n_p adjust slowly. This is the deterministic skeleton of the timing-search dynamics. Section 4 adds congestion damping and productivity shocks to the same skeleton, but the restoring-force coefficient φ/μ that pins the natural frequency in Theorem 1 is unchanged through both extensions.

Theorem 1 (Exact harmonic oscillations)**.**
In a timing search equilibrium (Definition 1) under Assumptions 1–3, and provided that entry occurs at all dates (the regularity condition formalized in the verification paragraph below, equivalent to $A \le \delta n_p/\sqrt{\omega^2+\delta^2}$), the equilibrium path is:

$$n(t) = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}. \tag{13}$$

The solution is exact within the feasible all-entry region and requires no linearization. The amplitude A and phase ϕ are determined by initial conditions (subject to the feasibility bound), and the ironing condition holds with $E = \frac{\varphi}{2}A^2$.

Proof. The ironing condition (Proposition 1) gives (11). Differentiation yields the harmonic oscillator equation (12), whose general solution is (13) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the Gaussian specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t+\phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t+\phi) = \frac{\varphi}{2}A^2 = E$. ∎

Verification and scope The pair $(n(t), e(t))$ is a timing search equilibrium for any amplitude $A \leq \delta n_p/\sqrt{\omega^2+\delta^2}$, the binding sufficient condition for $e(t) = \dot{n} + \delta n \geq 0$ throughout the cycle (Online Appendix B.6). Industries with net entry routinely $|\dot{n}| > \delta n_p/2$ are outside the regime where the ironing condition holds globally. The center structure leaves A free; the stochastic extension and dissipation (Propositions 2 and 3) select a unique stationary distribution from primitives disjoint from those pinning the period.

Structural decomposition The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator φ is the curvature of the profit landscape: how sharply profits fall when the industry deviates from optimal scale. The denominator μ is the timing friction: how costly it is to adjust the number of active firms rapidly, arising from congestion in the entry matching process. Treated as an abstract pair, (φ, μ) alone determines ω; the frequency does not depend on productivity z (which affects profit levels but not curvature), on the realized exit rate δ, or on the shock process (Section 4). When μ is calibrated structurally via Assumption 3, $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ inherits dependence on the matching elasticity α and the steady-state throughput δn_p; the resource-stock R underlying the matching technology is absorbed into the steady-state calibration of n_p and does not appear separately in the formula. The structural decomposition is therefore: ω depends on (φ, μ) as primitives, and on (α, δ, n_p) when μ is read off the microfoundation. Table 1 contrasts this decomposition with the way standard frameworks pin cycle frequency.

Multi-industry general equilibrium Because the wage is fixed and households have linear utility, each industry's profit landscape depends only on its own number of active firms. Under idiosyncratic shocks and no inter-industry input-output linkages or demand spillovers, a multi-industry economy with I industries each characterized by (φ_i, μ_i) evolves as I independent timing-search systems, with aggregate spectral density a superposition of industry-specific peaks at frequencies $\omega_i = \sqrt{\varphi_i/\mu_i}$.[4] The isolation of frequency from the shock process is therefore a *cross-sectional* prediction: different industries cycle at different frequencies determined by their respective market structures. Under common shocks or input-output linkages, the spectral superposition does not survive aggregation. Section 7 bounds the scope of the closed-form result, with quantitative leakage of (φ, μ) into the welfare cost under CRRA preferences documented in Online Appendix E.1.

Table 1: What determines cycle frequency?

	Frequency determined by	**Cross-industry prediction**
Canonical RBC	Shock autocorrelation	Common if shocks are common
Canonical NK	Policy rule	Common across industries
Beaudry et al. (2020)	Bifurcation condition	Not the object of their analysis
Timing search	$\boldsymbol{\omega = \sqrt{\varphi/\mu}}$	**Industry-specific frequency**

Notes: In canonical RBC and NK specifications, the spectral peak reflects the shock process or the policy rule—objects common across industries; multi-sector enrichments can produce dispersion but do not, by themselves, impose a γ-dependent slope (Online Appendix F.2). In Beaudry et al. (2020), the frequency emerges from a Hopf bifurcation; cross-industry frequency variation is not the object of their analysis. In timing search, the frequency is a closed-form function of two market-structure primitives (φ: profit-landscape curvature; μ: timing friction), is exact under linear utility, and is isolated from the shock process (Proposition 2).

3.5 Intuition: why the frequency is $\sqrt{\varphi/\mu}$

The closed-form result is the timing-search industry behaving like a spring.

[4] This superposition is a microfounded Fourier decomposition of the aggregate cycle. The ironing condition pins each unit's optimal entry path as the harmonic function $n_i(t) = n_{p,i} + A_i \cos(\omega_i t + \phi_i)$ at its own structural eigenfrequency $\omega_i = \sqrt{\varphi_i/\mu_i}$—the individual optimal decision is itself a Fourier basis element by construction, not a basis element imposed from outside. Pushed to a continuous heterogeneity index, the aggregate cycle $X(t) = \int n_i(t)\, dF(i)$ is the literal superposition of these basis elements weighted by the heterogeneity distribution F over (φ, μ), with bandwidth set by bottleneck damping (Section 4.4). The decomposition relies on three conditions: idiosyncratic shocks, no input-output linkages, and linear utility. With granular establishment- or region-level data, the empirical aggregate spectrum can be read as a structural deconvolution problem on F, an extension that current 3-digit cross-sectional tests do not exploit.

Why the industry oscillates: a spring with stiffness φ and inertia μ. As an industry expands toward optimal scale, firms keep entering because profits are above the steady-state norm. The ironing condition—no firm can profit by entering at a different date—pins entry to a smooth, equalized rate along the approach. That pinned rate is the catch: at n_p, entry cannot drop discontinuously to zero without creating arbitrage, so the industry overshoots. Beyond n_p, the same indifference logic drives entry to the other side of replacement, the flow reverses, and the industry oscillates rather than settling. The economics maps cleanly onto classical mechanics: profit curvature φ acts as a *restoring force* (steeper fall, harder pull back); the timing friction μ acts as *inertia* (larger μ, harder to change the pace of entry). A mass-spring system at $\omega = \sqrt{k/m}$ oscillates at the same frequency for the same reason as the timing-search industry at $\omega = \sqrt{\varphi/\mu}$. Both forces are necessary: take away the scale externality and the spring has no stiffness; take away the matching friction and the industry has no inertia, and oscillation collapses into a quiet return to optimum.

Isochrony: big and small cycles run on the same clock. The cycle's period does not depend on how hard the industry is hit. A deep recession leaves the industry far below optimal scale, profits per active firm are well above norm, and the ironing condition sustains a fast inflow; but the deeper trough means the industry has further to travel. A small shock generates weaker incentive, slower entry, and a shorter path. Speed and distance scale together, so $T = 2\pi\sqrt{\mu/\varphi}$ is the same in either case. Isochrony extends to the damped equilibrium: successive swings preserve their period even as the envelope decays (Online Appendix C.1). A direct corollary is the plucking pattern: deeper recessions recover faster (maximum return velocity $A\omega$ scales linearly with amplitude) but in the same time, the timing-search analogue of Friedman (1993); Bordo and Haubrich (2017); Dupraz et al. (2019) tested in Section 5.4.

3.6 Properties of the cycle

Corollary 1 (Properties of the deterministic cycle)**.** *Under Theorem 1:*

(i) Comparative statics. $\partial\omega/\partial\varphi > 0$ *and* $\partial\omega/\partial\mu < 0$*.*

(ii) Period. $T = 2\pi\sqrt{\mu/\varphi}$*, depending only on* (φ, μ)*.*

(iii) Amplitude invariance. *ω is independent of the cycle amplitude A and of productivity z, which scales profits without changing curvature.*

(iv) Free amplitude. *The deterministic system is a center: orbits form a one-parameter family indexed by the equilibrium value level E, and the deterministic dynamics do not select among them. The stochastic extension (Proposition 3) selects a unique stationary distribution.*

Proof. Items (i)–(iii) follow by direct calculation from $\omega = \sqrt{\varphi/\mu}$ in Theorem 1, which contains no dependence on (z, A). Item (iv) is the planar Hamiltonian property of the frictionless law of motion (12): in the absence of dissipation, $\frac{\mu}{2}\dot{n}^2 + \frac{\varphi}{2}(n - n_p)^2 = E$ is conserved along orbits, so each value of $E \geq 0$ traces a distinct closed orbit and the dynamics fix the period but not the amplitude. ∎

Beyond the Gaussian The closed-form result extends to general hump-shaped profits: under any C^2 profit function π with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, the frequency takes the form $\omega = \sqrt{|\pi''(n_p)|/[\mu\,\pi(n_p)]} + O(A^2)$, and the Gaussian-based prediction $T_0 = 2\pi\sqrt{\mu/\varphi}$ stays within 1.6% of the exact orbit-integrated period at $A/n_p = 0.30$ across 8 smooth hump-shaped profit specifications (Online Appendix B.2 and C.2). Exact isochrony is the $c_4 = 0$ knife-edge of the finite-amplitude expansion $\omega(A) = \omega_0(1 + c_4 A^2 + O(A^4))$, which the Gaussian satisfies (Online Appendix B.4). Relaxing the Gaussian's exact symmetry generates the empirically standard asymmetry in expansions and contractions: when $\pi'''(n_p) < 0$, the economy spends more time below n_p (slow expansions, sharp contractions), with the asymmetry parameter pinned by $\pi'''(n_p)/|\pi''(n_p)|$ (Online Appendix B.4).

4 Bottleneck Friction and the Stationary Cycle

The frictionless cycle of Theorem 1 is a center: a one-parameter family of closed orbits with free amplitude. Two ingredients are needed to convert this skeleton into the equilibrium that confronts the data, and each contributes a distinct empirical signature. *Bottleneck friction* (Section 4.1) on its own dissipates cycle amplitude over time while preserving the structural period—the plucking / manufacturing-echo signature documented in Section 5.4. *Stochastic shocks* (Section 4.3) on their own preserve the spectral peak at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$, but the amplitude diffuses without bound. Together (Section 4.4) they yield the damped stochastic oscillator at stationary amplitude $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$. Section 4.5 closes with the frequency-welfare separation that the stationary cycle delivers in closed form.

4.1 Bottleneck friction

Assumption 4 (Bottleneck friction).
The supplier-side resource flow contributes an additive quadratic congestion cost $\frac{1}{2}\chi\,\dot{n}^2$ with $\chi > 0$, generating a damping force $-\chi\dot{n}$ in the equilibrium equation of motion via the Lagrangian dissipation function $\mathcal{R}(\dot{n}) = \frac{1}{2}\chi\dot{n}^2$.[5]

What the bottleneck friction captures, and how it differs from μ. Although both summarize congestion, μ and χ sit on opposite sides of the entry process and play different roles in equilibrium. The matching friction μ is on the *firm side*—the matching cost each entering firm pays inside the kernel $\xi(\dot{n}) = e^{-(\mu/2)\dot{n}^2}$, microfounded as $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ from Cobb-Douglas matching (§2.3)—and acts as inertia in the equilibrium law of motion, pinning the cycle's frequency. The bottleneck friction χ is on the *supplier side*—the convex resource cost the supplier sector pays to deliver net adjustment $\dot{n}$ at peak strain, microfounded as $\chi = c_2 r^2$ from the bottleneck curvature c_2 and the resource-flow linkage r (Online Appendix B.5)—and acts as dissipation, pinning the cycle's amplitude. The disjoint determinants of frequency ($\omega_0 = \sqrt{\varphi/\mu}$) and welfare ($\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$) in Theorem 2 follow directly from this separation of mechanism: μ is paid inside firms' profits, χ is paid by households for supplier-side infrastructure.

4.2 Deterministic damped case: amplitude decays, period stays

Before adding shocks, bottleneck friction acting alone already delivers a distinctive cycle property that the data can verify. Setting $\sigma_\varepsilon = 0$ in the equilibrium law of

[5] The dissipation function $\mathcal{R}(\dot{n}) = \frac{1}{2}\chi\dot{n}^2$ is known in classical mechanics as the Rayleigh dissipation function. We adopt "bottleneck friction" for χ throughout the body to signal its economic role as a supplier-side resource bottleneck in the entry pipeline, and reserve "Rayleigh" for the formal Lagrangian derivation in Online Appendix B.5.

motion with bottleneck friction:

$$\mu\,\ddot{n} \;+\; \chi\,\dot{n} \;+\; \varphi(n - n_p) \;=\; 0. \tag{14}$$

For an industry displaced from n_p by a one-time shock—an initial deviation A_0 with zero initial velocity—the equilibrium path is

$$n(t) - n_p \;=\; A_0\, e^{-\eta t/2}\, \cos(\omega_d\, t), \qquad \omega_d \equiv \sqrt{\omega_0^2 - \eta^2/4}, \quad \eta \equiv \chi/\mu,$$

where the speed at which bottleneck friction dissipates output is $\eta = \chi/\mu$, the ratio of bottleneck friction to the matching friction's inertial weight. Three properties follow.

- *The period barely changes.* At empirical damping ($\eta^2/(4\omega_0^2) \lesssim 5\%$) the cycle length stays essentially at the structural $T = 2\pi/\omega_0$ throughout the decay, falling at most $O(\eta^2)$ below it.
- *The amplitude shrinks toward steady state.* Each successive swing is smaller than the last by a factor that depends on the ratio of bottleneck friction χ to matching friction μ; the amplitude does not diverge.
- *Period and amplitude decay are independent.* A heavier bottleneck shrinks amplitude faster but barely changes the cycle length, in contrast to AR(1) and similar linear-decay models where persistence and period are mechanically linked.

This is the *plucking / echo pattern*: a one-time shock triggers oscillations that lose amplitude over time while keeping their period. U.S. manufacturing investment after the Great Depression / WWII oscillates at $T = 5.0$ years for four decades while the amplitude decays (Section 5.4); oil and mining investment after 1979 oscillates at $T = 3.25$ years. The pattern distinguishes bottleneck dissipation from linear-decay alternatives such as AR(1), where decay rate and period are mechanically linked (Online Appendix C.1 plots the impulse responses).

4.3 Adding shocks

Consider an i.i.d. disturbance to profits:

$$\tilde{\pi}(n,t) = \pi(n) + \sigma_\varepsilon\, \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0,1). \tag{15}$$

The ironing condition no longer holds exactly: the equilibrium value level E drifts stochastically with the shock, and the economy migrates between orbits. Without bottleneck friction this migration is unbounded—the cycle's amplitude grows without bound on average and no stationary distribution exists—but the spectral peak remains at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$.

Proposition 2 (Stochastic limit cycle)**.**
Under small shocks (15)*:*

(i) The spectral density of $n(t)$ peaks at the structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ in the undamped small-shock limit; with bottleneck damping $\eta = \chi/\mu > 0$, the observed PSD peak is $\omega_{PSD} = \sqrt{\varphi/\mu - \eta^2/2}$, equal to ω_0 up to an $O(\eta^2)$ correction (Proposition 3(iv)).

(ii) The peak frequency is determined by (φ, μ) at leading order and by the damping ratio η at second order, not by σ_ε.

(iii) Even i.i.d. shocks produce persistent cycles: the persistence is endogenous to the equilibrium structure.

Proof. See Online Appendix B.1. ■

The structural frequency is a property of market structure, not the shock process: i.i.d. shocks produce a PSD peak at $\omega_{\text{PSD}} = \sqrt{\omega_0^2 - \eta^2/2}$, while persistent shocks shift the peak negligibly when the transfer function is sharp. This matches the central finding of Beaudry et al. (2020)—persistent boom-bust dynamics from short-lived disturbances—though the mechanism here is the scale-congestion trade-off rather than financial-friction complementarities; numerical confirmation is in Online Appendix C.1.

4.4 The stationary cycle: damped stochastic oscillator

In the stationary cycle—with both bottleneck dissipation and profit shocks—the industry oscillates at the same natural frequency $\omega = \sqrt{\varphi/\mu}$ but with a well-defined stationary amplitude: dissipation pulls the industry toward optimal scale, shocks re-excite it, and the balance pins the variance of $(n - n_p)$ at $\sigma_\varepsilon^2/(2\chi\varphi)$ (Proposition 3). Differentiating the ironing condition and balancing the value-erosion rate against the bottleneck's output-dissipation rate $\chi\dot{n}^2$ yields the equation of motion (Online Appendix B.5):

$$\underbrace{\ddot{n}}_{\substack{\text{change in}\\ \text{entry rate}}} + \underbrace{\eta\,\dot{n}}_{\substack{\text{congestion}\\ \text{damping}}} + \underbrace{\frac{\varphi}{\mu}\,(n-n_p)}_{\substack{\text{pull toward}\\ \text{optimal scale}}} = \underbrace{\frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t}_{\substack{\text{productivity}\\ \text{shock}}}\,, \tag{16}$$

with $\eta = \chi/\mu$. The natural frequency $\omega_0 = \sqrt{\varphi/\mu}$ is unchanged: bottleneck friction and shocks shape how severely the industry oscillates, not the period at which it does so.

Proposition 3 (Amplitude selection)**.**
With bottleneck friction coefficient $\chi > 0$ (damping $\eta = \chi/\mu > 0$) and shocks $\sigma_\varepsilon > 0$:

(i) Without shocks ($\sigma_\varepsilon = 0$): orbits spiral inward to $(n_p, 0)$ with decay half-life $\ln 2 \cdot 2\mu/\chi$.

(ii) With shocks: the system has a unique stationary distribution with variance:

$$\mathbb{E}\big[(n-n_p)^2\big] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\varphi\,\mu} = \frac{\sigma_\varepsilon^2}{2\,\chi\,\varphi}, \tag{17}$$

where the second equality uses $\eta = \chi/\mu$, so that the matching friction μ cancels.

(iii) The Hilbert envelope $A = \sqrt{(n-n_p)^2 + (\dot{n}/\omega_0)^2}$ follows a Rayleigh distribution with scale $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, so that $\mathbb{E}[A^2] = 2\sigma_x^2 = \sigma_\varepsilon^2/(\chi\varphi)$.

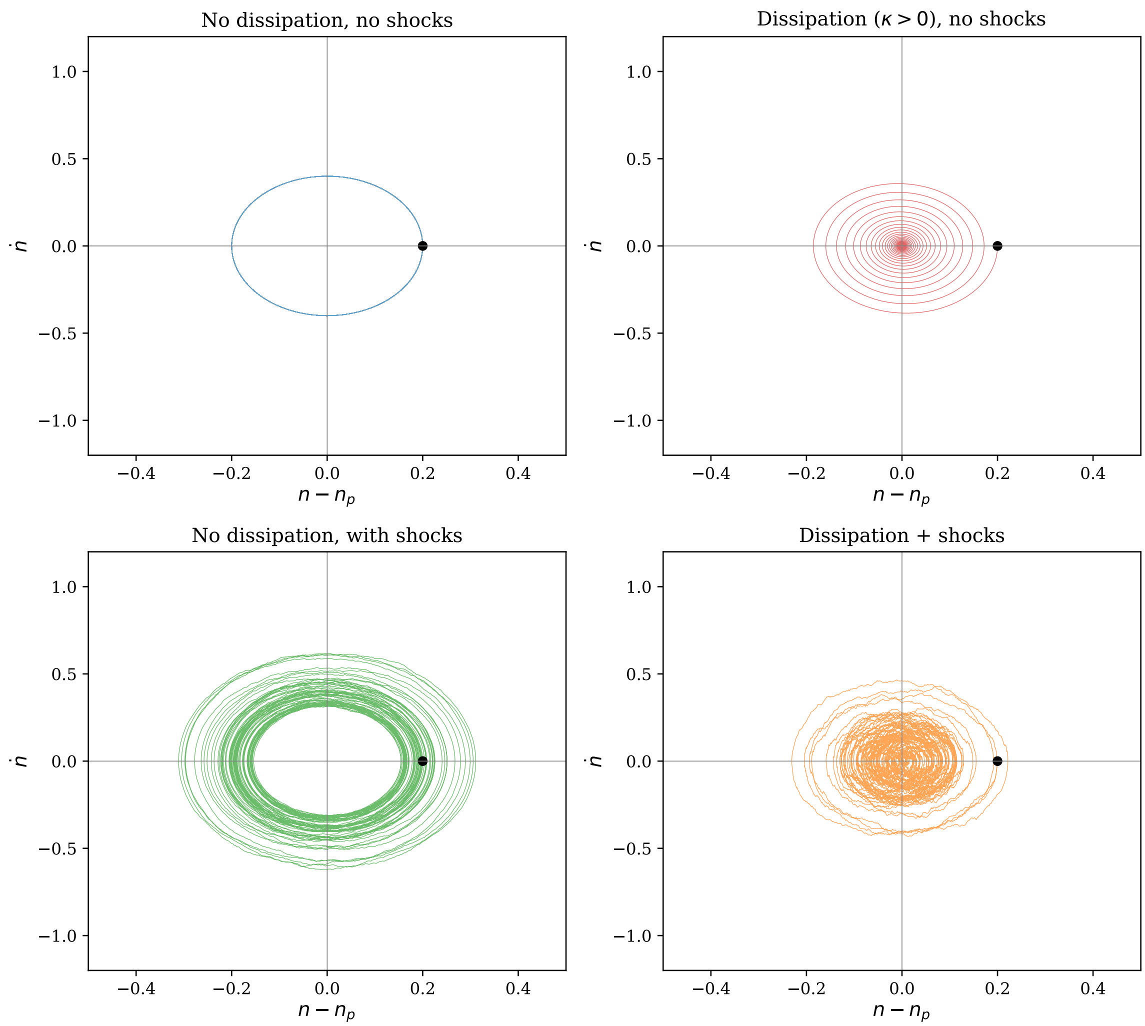
(iv) The spectral peak frequency of the driven oscillator's stationary spectrum is $\omega_{peak} = \sqrt{\varphi/\mu - \eta^2/2}$, obtained by maximizing the transfer function $|H(\omega)|^2 = 1/[(\omega_0^2 - \omega^2)^2 + \eta^2\omega^2]$ over ω. (The homogeneous damped natural frequency $\sqrt{\varphi/\mu - \eta^2/4}$, at which the unforced solution oscillates, is a separate object.) For small η, both quantities are negligibly below $\sqrt{\varphi/\mu}$.

Proof. See Online Appendix B.5. ∎

Figure 1 visualizes the four dissipation/shock regimes: dissipation alone makes orbits spiral inward to n_p (top right), shocks alone make orbits diffuse without bound across amplitudes (bottom left), and only with both ingredients (bottom right) does the balance pin a stationary distribution at $\mathbb{E}[(n-n_p)^2] = \sigma_\varepsilon^2/(2\chi\varphi)$. Online

Appendix C.1 confirms numerically that the Hilbert envelope follows the Rayleigh density of Proposition 3(iii).

Figure 1: Amplitude selection through bottleneck dissipation



Notes: Phase portraits $(n - n_p, \dot{n})$ under four regimes. Top left: deterministic orbit (closed ellipse, free amplitude). Top right: dissipation without shocks (inward spiral to steady state). Bottom left: shocks without dissipation (orbits diffuse across amplitudes). Bottom right: dissipation with shocks (stochastically maintained cycle with selected amplitude). The balance of dissipation and forcing pins $\mathbb{E}[(n - n_p)^2] = \sigma_\varepsilon^2/(2\eta\varphi\mu)$.

Substituting $\eta = \chi/\mu$ gives $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, in which μ cancels exactly: industries with volatile profits, low bottleneck friction, or flat profit landscapes experience larger cycles, while μ governs only the cycle frequency. Combining with the welfare formula (21):

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\,\chi}. \tag{18}$$

Higher bottleneck friction χ *reduces* welfare loss by damping the cycle—a "stabilization through friction" result—with the welfare cost depending only on $(\sigma_\varepsilon, \chi)$ as φ cancels.

4.5 Frequency-welfare separation

The closed-form structural frequency $\omega_0 = \sqrt{\varphi/\mu}$ (Theorem 1), the stationary variance $\sigma_x^2 = \sigma_\varepsilon^2/(2\chi\varphi)$ (Proposition 3 after substituting $\eta = \chi/\mu$, so that μ cancels), and the welfare cost $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ (Section 6.1) together establish the paper's central structural claim: *structural frequency and welfare cost have disjoint determinants*, with cycle severity standing as an intermediate observable depending on the overlapping set $(\sigma_\varepsilon, \chi, \varphi)$. We state this as a single theorem.

Theorem 2 (Frequency-welfare separation)**.**
In the timing search equilibrium under Assumptions 1–4 and the damped-stochastic specification of Proposition 3, with $\eta = \chi/\mu$:

$$\begin{aligned}
&\textit{structural frequency:} && \omega_0 = \sqrt{\varphi/\mu} && \textit{depends only on } (\varphi, \mu);\\
&\textit{cycle severity:} && \sigma_x = \sigma_\varepsilon/\sqrt{2\,\chi\,\varphi} && \textit{depends on } (\sigma_\varepsilon, \chi, \varphi);\\
&\textit{welfare cost:} && \mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi) && \textit{depends only on } (\sigma_\varepsilon, \chi).
\end{aligned}$$

The structural frequency depends on (φ, μ); the welfare cost depends on $(\sigma_\varepsilon, \chi)$; these determinant sets are disjoint, with cycle severity the intermediate observable connecting both. The structural frequency is invariant to σ_ε, χ, the shock persistence (in the i.i.d. specification), and the realized cycle amplitude; the welfare cost is invariant to (φ, μ) and to the per-firm entry cost κ. The observed PSD peak inherits the structural frequency up to an $O(\eta^2)$ correction: $\omega_{PSD} = \sqrt{\varphi/\mu - \eta^2/2}$ (Proposition 3(iv)).

Proof. The structural-frequency claim is Theorem 1 (deterministic linearized dynamics admit eigenfrequency $\omega_0 = \sqrt{\varphi/\mu}$, depending on (φ, μ) alone). The severity claim follows by substituting $\eta = \chi/\mu$ into Proposition 3(ii): $\sigma_x^2 = \sigma_\varepsilon^2/(2\eta\varphi\mu) = \sigma_\varepsilon^2/(2\chi\varphi)$, so μ cancels exactly. The welfare claim follows from $\mathcal{L} \approx (\varphi/2)A^2$ in (21) and the Rayleigh-envelope identity $\mathbb{E}[A^2] = 2\mathbb{E}[(n - n_p)^2] = \sigma_\varepsilon^2/(\chi\varphi)$; substituting gives

$\mathbb{E}[\mathcal{L}] = (\varphi/2)\mathbb{E}[A^2] = \sigma_\varepsilon^2/(2\chi)$, with the dependence on φ canceling. The observed-PSD-peak claim is Proposition 3(iv). ■

The theorem has three methodological consequences. First, cross-industry variation in ω_0 identifies the market-structure ratio φ/μ free of contamination from χ, σ_ε, or κ (the $O(\eta^2)$ PSD-peak shift is below 5% at empirical damping); the cross-sectional regression of Section 5.3 therefore tests a structural prediction about φ/μ even when proxies for the individual primitives are imperfect. Second, two industries with identical (φ, μ) but different shock environments cycle at the same frequency, making the cross-industry frequency prediction structural rather than a reduced-form artifact of common shocks. Third, industrial policy targeting μ (entry regulation) or φ (cluster development) affects timing alone; only policies that move σ_ε or χ affect welfare (Section 6.2).

Scope of the separation Theorem 2's welfare claim is exact under linear utility, the per-firm effective-output welfare metric of (21), and exclusion of the supplier's $\frac{1}{2}\chi\dot{n}^2$ resource flow; relaxing each condition reintroduces leakage of (φ, μ) into the welfare cost of order 10–25% at the median under standard CRRA preferences or alternative metrics, documented in Online Appendix E.1.

Comparison to alternative endogenous-cycle models Timing search and Beaudry et al. (2020) share the broader insight that endogenous structure can turn short-lived disturbances into persistent cyclical adjustment, but Beaudry et al. (2020) generate a stochastic limit cycle through financial frictions and a Hopf bifurcation, while timing search generates characteristic frequencies from the entry-date indifference condition and a scale–congestion trade-off; the latter mechanism, alone, delivers the industry-decomposable closed form $\omega_0 = \sqrt{\varphi/\mu}$ tested in Section 5.3. Across standard benchmarks (AR(1) RBC, NK AR(2), Van der Pol, multi-sector RBC with input-output linkages, and Khan and Thomas (2008)-style heterogeneous-θ adjustment costs) the simulated cross-industry slope $\hat{\beta}_\gamma$ centers near zero with 95% CIs roughly $[-0.13, +0.20]$, none of 50 replications reaching the empirical $[0.41, 0.48]$ range; the comparison clarifies what additional structure the empirical question requires rather than ruling out enriched versions of these models (Online Appendices F.1–F.4).

5 Quantitative Framework

The model predicts a characteristic frequency, not regular calendar-time cycles. Real industries are hit by many shocks, and observed cycles are noisy and intermittent; after sufficiently large disturbances, however, the response should contain a frequency component whose location is governed by market-structure primitives. The empirical analysis therefore asks whether industry fluctuations contain systematic frequency differences associated with the model's primitives. Theorem 1's closed form $\omega = \sqrt{\varphi/\mu}$ implies a log-linear cross-industry regression with structural slope $1/2$ on $\ln \varphi$; under matching friction approximately uniform within manufacturing, this is what the data should reveal in the cross-section.

The model delivers three testable cross-industry predictions. First, cycle frequency should vary with market-structure proxies, not with the shock process. Second, large one-time shocks should trigger deterministic oscillations at each industry's structural frequency, not at a common frequency inherited from the shock. Third, industries displaced by a common shock should overshoot their pre-shock trend rather than converge monotonically. Section 5.3 tests the first prediction using the Ellison-Glaeser concentration index across 16 manufacturing 3-digit industries, with a parallel cross-section across 10 BDS subsectors using independently measured spatial concentration as a robustness extension. Section 5.4 tests the second and third predictions through three diagnostics: post-shock periodicity in manufacturing and oil investment using Fisher's g-test, BDS net-entry oscillation features across 19 sectors, and the post-COVID overshoot.

5.1 Cross-industry predictions

Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields the cross-industry regression framework

$$\ln \omega_i = \tfrac{1}{2} \ln \varphi_i - \tfrac{1}{2} \ln \mu_i + u_i, \tag{19}$$

with the slopes $\pm 1/2$ as *exact* model predictions. The headline test of Section 5.3 narrows to 16 manufacturing 3-digit industries, within which the matching technology (permits, specialist construction labor, equipment supply chains) is broadly common and turnover varies only 1.7×; under approximate μ uniformity, the prediction collapses to a φ-driven slope $\beta_\gamma = 1/2$. Two industries with identical shock processes but different (φ, μ) cycle at different frequencies—the prediction that distinguishes timing search from RBC and NK frameworks, where the spectral peak mirrors the shock process or the policy rule.

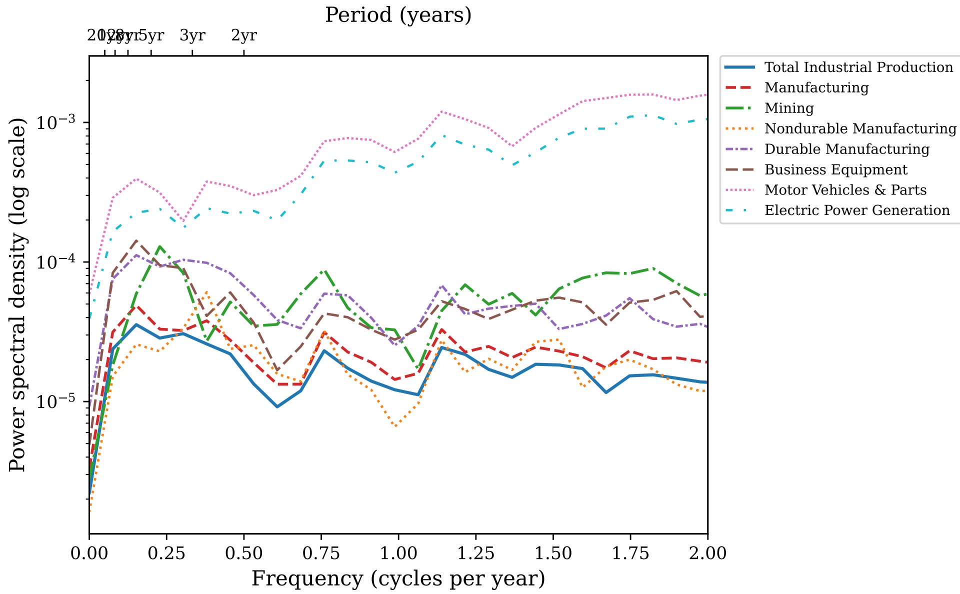
5.2 Spectral evidence from U.S. industry data

I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Figure 2 plots the power spectral density of log-differenced production for each industry. Two features stand out. First, spectral peaks vary substantially across industries. Motor Vehicles, Durable Manufacturing, Business Equipment, and aggregate Manufacturing concentrate spectral mass near periods of 6–7 years. Mining peaks at 4.4 years. Nondurable Manufacturing and Electric Power peak near 2.6 years. Second, the cross-industry clustering is informative: industries sharing supply-chain linkages—durables, motor vehicles, business equipment—cluster at similar frequencies despite producing very different goods. This is consistent with ω depending on market structure, not on product-specific shocks.

Figure 3 summarizes the dominant cycle period for each industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). The cross-industry dispersion is substantial: dominant periods range from 2.6 years (Electric Power, Nondurable Manufacturing) to 6.6 years (Durable Manufacturing, Motor Vehicles). This two-and-a-half-fold variation in cycle frequency across industries that share a common macroeconomic environment—the same monetary policy, the same aggregate demand shocks, the same financial conditions—is difficult to reconcile with models that derive spectral properties from a common shock process. In the timing search framework, the variation is structural: it reflects cross-industry differences in market-structure primitives. Across this broad eight-industry sample—which spans Manufacturing, Mining, and Electric Power—both the profit-landscape curvature (φ_i) and the timing friction (μ_i) plausibly vary across sectors with different matching technologies. The headline test of Section 5.3 narrows to 16 manufacturing 3-digit industries, within which μ is approximately uniform, so

Figure 2: Spectral density of U.S. industry production growth



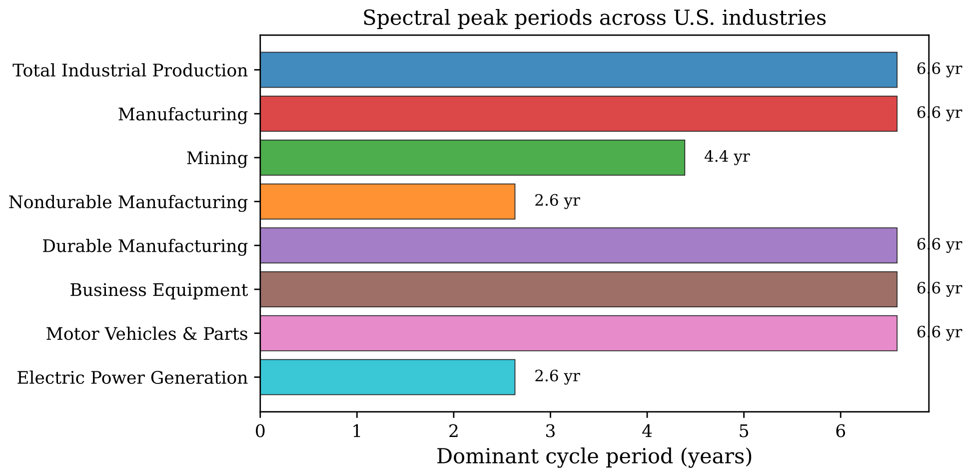
Notes: Power spectral density (Welch's method) of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12. The top axis labels convert frequency (cycles per year) to period (years). Industries with strong supply-chain linkages (Motor Vehicles, Business Equipment, Durable Manufacturing) cluster near $T \approx 6$–7 years, while industries with different market structures (Mining, Nondurables, Electric Power) show distinct peak frequencies.

the cross-industry signal collapses to the φ channel.

Model shape versus data shape The timing-search SDE generates an internal spectral peak at $\omega_0 = \sqrt{\varphi/\mu}$, while an AR(1) on growth rates produces a smooth low-pass spectrum with no internal peak. Figure 4 compares the model PSD at a manufacturing-like calibration ($T_0 = 5.0$ years) to the FRED industrial production PSD for NAICS 337 (Furniture), which peaks at $T = 4.9$ years: the model can produce a shape comparable to the data at a single industry, providing qualitative motivation for the cross-industry slope test below.

Consistency check: nonlinear dependence in industry dynamics As a consistency check rather than a discriminating test, we apply the BDS test (Brock et al., 1996) to AR(1)-filtered industry residuals. The i.i.d. null is rejected at the 1% level for all eight industries, with z-statistics ranging from 16.2 (Durable Manufacturing) to 21.8 (Motor Vehicles & Parts). This indicates nonlinear or nonstationary structure

Figure 3: Cross-industry variation in spectral peak periods



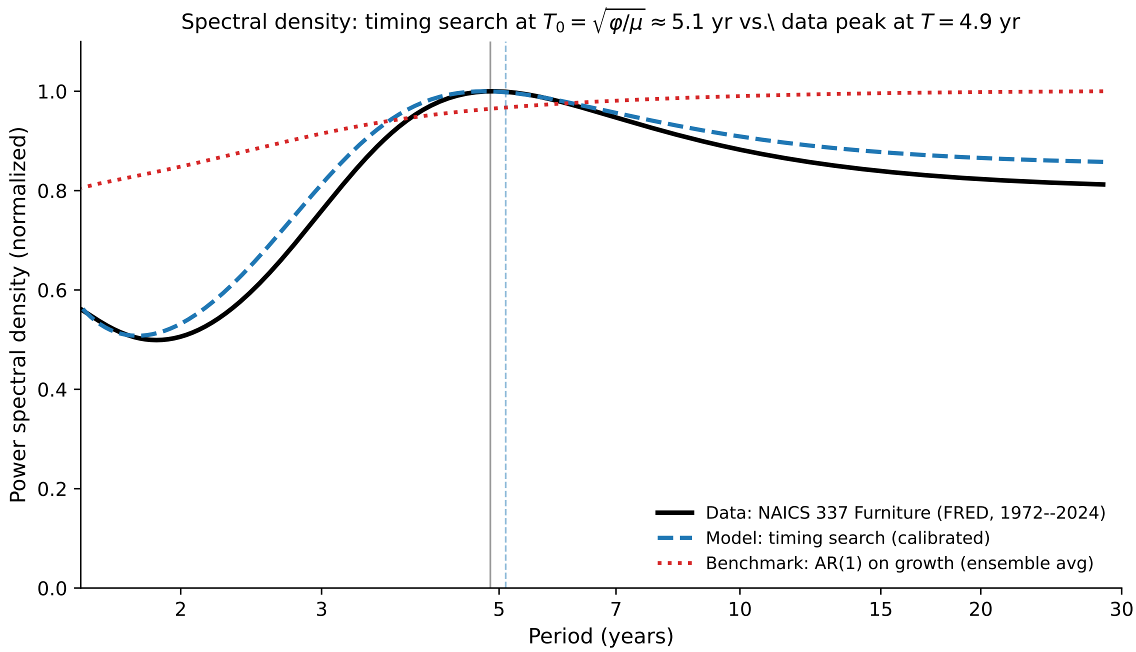
Notes: Dominant cycle period (years) for each U.S. industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). Industries with strong scale–congestion externalities (Motor Vehicles, Durable Manufacturing, Business Equipment) cluster near $T \approx 6$–7 years. Industries with weaker or different externality structures (Mining, Nondurables, Electric Power) cycle at distinct frequencies.

beyond AR(1) in industry dynamics. The pattern is consistent with the timing-search mechanism but also with GARCH, regime switching, and structural breaks; the 1972–2024 sample spans the Great Moderation and COVID, both of which can produce comparable BDS rejection magnitudes. The BDS exercise therefore confirms departure from linear AR(1) dynamics but does not discriminate among the nonlinear alternatives. Full results across embedding dimensions 2–6, with AR(1) and AR(p)+GARCH pre-filtering specifications, are in Online Appendix D.2.

5.3 Cross-industry regression with external proxies

A baseline regression on seven broad industry groups using BEA proxies (inverse I-O self-purchase ratio for φ and capital per employee for μ) gives a univariate slope of 0.85 ($R^2 = 0.65$, $p = 0.028$) on $\ln \hat{\varphi}_i$ and a multivariate R^2 of 0.75, motivating the broad-sector relationship (Online Appendix D.1). The headline test scales this up: $N = 16$ three-digit manufacturing industries with the Ellison and Glaeser (1997) geographic concentration index γ_i as the φ proxy and Burg AR(24) spectral peaks as

Figure 4: Spectral density: timing-search model vs. industry data



Notes: Burg AR(24) power spectral densities, normalized to the maximum within the business-cycle band (1.5–30 years). Solid black: FRED monthly industrial production growth for NAICS 337 (Furniture), 1972–2024, with empirical peak at $T = 4.9$ years. Dashed blue: simulated timing-search stochastic differential equation at a manufacturing-like calibration ($\varphi = 1.52$, $\mu = 1$, $\kappa = 0.32$, $\sigma_\varepsilon = 0.04$, theoretical $T_0 = 2\pi/\sqrt{\varphi/\mu} = 5.1$ years; empirical Burg-AR peak at $T = 4.8$ years), ensemble-averaged over 30 replications, with realistic-conditions noise components (idiosyncratic measurement noise $\sigma_{\text{meas}} = 0.060$ and a slow AR(1) productivity-drift component, $\rho = 0.93$, $\sigma = 0.0011$) that mimic the finite-sample data generating process. Dotted red: ensemble average of 30 AR(1)-on-growth-rate replications ($\rho = 0.5$), shown for comparison—a standard "no internal peak" benchmark with smooth low-pass spectrum. Timing search can generate a comparable peak and surrounding spectral shape; AR(1) cannot.

ω_i.[6]

Table 2 reports the results. In the univariate specification, the Ellison and Glaeser (1997) concentration index significantly predicts cycle frequency ($\hat{\beta}_\gamma = 0.20$, $p = 0.010$, $R^2 = 0.39$, $N = 16$). The sign matches the theory: geographically concentrated industries—those with stronger agglomeration externalities (higher φ)—cycle faster.

Two refinements close most of the apparent gap to the theoretical slope $\beta_\gamma = 0.5$.

Alternative spectral estimators. A multitaper PSD (Thomson DPSS, $K = 5$ tapers)

[6]The proxy swap is empirical: at the 3-digit level, the broad-sector I-O and capital-intensity proxies do not preserve the relationship; the EG index, constructed at the 3-digit manufacturing level by design, does. Online Appendix D.1 documents.

gives $\hat{\beta}_\gamma = 0.41$ ($p = 0.004$, $R^2 = 0.47$), and Burg AR(12) gives $\hat{\beta}_\gamma = 0.48$ ($p = 0.029$). The Burg AR(24) baseline with $\hat{\beta}_\gamma = 0.20$ is the most conservative point estimate; multitaper and shorter-order Burg both deliver coefficients within one standard error of the theoretical 0.5 (Online Appendix D.6 reports the full six-estimator panel).

2SLS with the 1947 Hoover instrument. Instrumenting the modern γ_i with the 1947 Hoover localization coefficient from Kim (1995) yields a 2SLS estimate of $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.45$ (HC1 SE 0.17, $N = 15$), with first-stage $F = 5.30$ ($\pi_{1947} = 1.21$, HC1 SE 0.53, first-stage $R^2 = 0.27$). Standard Wald (1943) inference rejects $\beta_\gamma = 0$ at $p = 0.020$ and fails to reject the theoretical $\beta_\gamma = 0.5$ at $p = 0.75$; Anderson and Rubin (1949) weak-IV-robust testing rejects $\beta_\gamma = 0$ at $p = 0.007$ and fails to reject $\beta_\gamma = 0.5$ at $p = 0.78$. The first-stage F is below the conventional Stock and Yogo (2005) weak-IV threshold, so we read the IV as corroborative rather than independently identifying.

We use 1947 as the headline instrument because it is the earliest Kim year for which all NAICS-3 manufacturing industries in the cross-section were established as distinct categories with nonzero Hoover loading; many 3-digit categories were not yet distinct in 1900, giving the 1900 coefficient near-zero first-stage loading across these industries. As a robustness check, the joint 2SLS using all three Kim years (1900, 1927, 1947) yields essentially the same $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.45$ with Hansen (1982) overidentification $p = 0.96$.

Pre-1947 industrial geography pre-dates the modern macroeconomic environment in which cycle frequencies are observed, motivating the exclusion restriction; we acknowledge the standard caveat that historical concentration could correlate with persistent industry traits that affect modern cycle frequency through channels other than current γ. The IV result is best interpreted alongside the OLS, multitaper, and BDS subsector estimates as a coherent body of evidence pointing to β_γ in the neighborhood of the theoretical 1/2.

Structural reading of the slope. The model predicts $\omega_i = \sqrt{\varphi_i/\mu_i}$, so the regression slope on $\ln \gamma_i$ is 1/2 only if $\ln \mu_i$ is approximately uncorrelated with $\ln \gamma_i$ across the cross-section. Within manufacturing, this is plausible on first principles: the matching technology (permits, specialist construction labor, equipment supply chains) is a generic property of the entry-matching pipeline rather than an industry-specific friction, and the structural formula $\mu_i = (1-\alpha)/[\alpha(\delta_i n_{p,i})^2]$ produces narrow within-manufacturing variation when α is approximately uniform and turnover δ_i ranges only $1.7\times$ ($\delta \in [0.065, 0.11]$). The empirical recovery of $\beta_\gamma \approx 1/2$ is therefore consistent with

μ approximately uniform across manufacturing industries: cross-industry frequency variation operates through the profit-landscape curvature φ, which has an order-of-magnitude wider range (γ varies from 0.004 to 0.12, a 30× range).

External μ proxies that mix friction with industry size, asset durability, or regulatory burden are wrong-signed (Online Appendix D.5), consistent with the interpretation that the friction-relevant component of μ is not what those proxies measure. The economic content of approximate μ-uniformity within manufacturing is that input factors relevant to the entry-matching technology—permits, specialist construction labor, equipment supply chains, contractor networks—flow flexibly across 3-digit manufacturing industries; what differentiates them is not their access to entry resources but the agglomeration density and supplier-network structure they enter into.

Table 2: Structural regression: geographic concentration and cycle frequency

	OLS $N = 16$	OLS multitaper $N = 16$	2SLS (historical IV) $N = 15$	Theory
$\hat{\beta}_\gamma$	0.20	0.41	0.45	0.50
HC1 SE	(0.07)	(0.12)	(0.17)	—
Wald p vs. 0	0.010	0.004	0.020	—
Wald p vs. 0.5	< 0.001	0.46	0.75	—
AR p vs. 0	—	—	0.007	—
AR p vs. 0.5	—	—	0.78	—
R^2 / 1st-stage F	0.39	0.47	$F = 5.30$	—

Notes: Cross-industry regression of $\ln \omega_i$ on $\ln \gamma_i$ (Ellison and Glaeser 1997 geographic concentration index). Column 1: OLS with Burg AR(24) spectral peaks (paper baseline). Column 2: OLS with multitaper spectral peaks (Thomson DPSS, $K = 5$, Thomson 1982). Column 3: 2SLS with the 1947 Hoover localization coefficient (Kim 1995) as the single instrument for $\ln \gamma_i$ (first-stage $\pi_{1947} = 1.21$, HC1 SE 0.53, $R^2 = 0.27$). Spectral peaks estimated from log-differenced monthly industrial production (FRED, 1972–2024). HC1 heteroskedasticity-robust standard errors (White, 1980). AR rows report Anderson and Rubin (1949) weak-IV-robust p-values, which apply to column 3 only. The first-stage $F = 5.30$ is below the conventional Stock and Yogo (2005) weak-IV threshold; column 3 should be read as corroborative of the structural slope rather than independently identifying. As a robustness check, the joint 2SLS with all three Kim years (1900, 1927, 1947) yields $\hat{\beta}_\gamma^{\mathrm{IV}} = 0.45$ (HC1 SE 0.16) with Hansen (1982) overidentification $p = 0.96$ but a weaker joint first-stage $F = 1.55$ (driven down by the near-zero first-stage loading on 1900).

Three robustness exercises further support the cross-sectional relationship without overturning it. First, a parallel $N = 10$ cross-section across BDS subsectors with clear interior spectral peaks, using BDS-derived spatial concentration G_i rather than the published γ_i, yields $\hat{\beta} = 0.107$ (HC1 SE 0.043, $p = 0.037$, $R^2 = 0.39$): same sign, similar R^2, different data source for both ω_i and the concentration measure (Online Appendix D.7; sample selection on a feature of the dependent variable, exploratory only). Second, a pooled US+UK cross-section using UK ONS Index of Production data (1990–2024) for 12 NACE-2 manufacturing divisions matched to the US 16 yields a pooled slope $\hat{\beta}_\gamma = 0.13$ ($p = 0.026$) with country fixed effects; the UK-only slope is small and insignificant, reflecting the strong assumption that US γ values approximate UK industry-specific concentration (UK BRES-derived γ would address this but requires non-trivial data construction; Online Appendix D.7). Third, at four-digit resolution ($N = 86$), Ellison et al. (2010) publish γ only at the three-digit level, so every four-digit child inherits its parent's γ and the slope mechanically collapses to $\hat{\beta}_\gamma = 0.027$ ($p = 0.705$); this tests the inheritance of γ rather than its predictive power at four-digit resolution (Online Appendix D.4).

Identification The ratio φ/μ is identified directly from the spectral peak ($\varphi/\mu \approx (2\pi/T^*)^2$ at leading order, with an $O(\eta^2)$ correction). Individual φ and μ require external information; χ is identified from $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$ given $(\sigma_\varepsilon, \varphi)$; and κ enters only the steady-state ironing level.

5.4 Additional diagnostics

The cross-sectional regression in Section 5.3 is the paper's structural test, identifying the φ channel through the geographic-concentration proxy. We complement it with five additional diagnostics that are consistent with the timing-search mechanism but rest on weaker identification: cross-industry decoupling of cycle period and cycle amplitude, oscillatory features in BDS net entry rates that lie outside the AR(1) class, post-shock periodicity in two historical investment series at industry-specific structural frequencies, an aggregate post-COVID overshoot pattern, and a recovery-speed-versus-recession-depth pattern in U.S. industrial production consistent with the model's plucking corollary. We present these as suggestive rather than definitive evidence; their joint pattern—interior spectral peaks across sectors, period-amplitude decoupling, post-shock periodicity at industry-specific frequencies, trend overshoot after a common displacement, and amplitude-proportional recovery speed—is harder to reproduce in any single alternative model than in any one diagnostic taken alone.

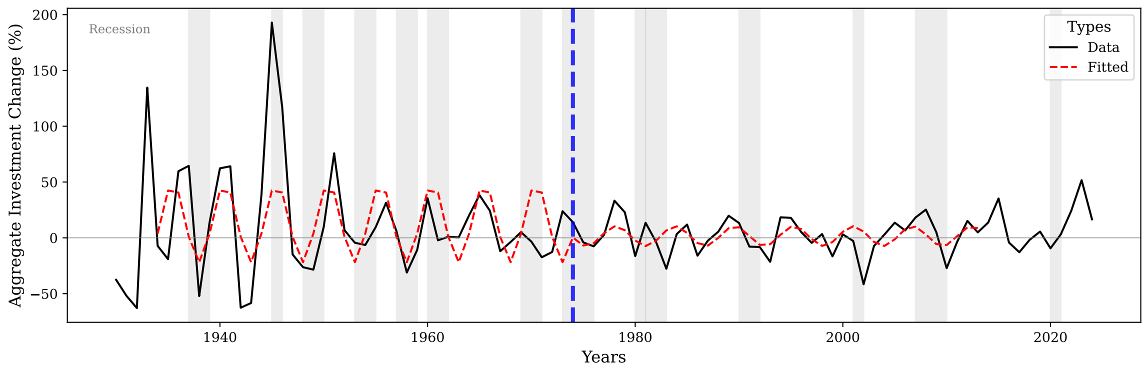
Period-amplitude decoupling in the cross-section Under μ-uniformity within manufacturing, the cycle period depends on φ_i alone while amplitude additionally depends on $(\sigma_{\varepsilon,i}, \chi_i)$ orthogonal to φ_i, predicting negligible cross-industry correlation between $\log T_i$ and $\log \sigma_{x,i}$. The data give Pearson $r = +0.21$ ($p > 0.3$, $N = 16$): period and amplitude are statistically uncorrelated, against the positive correlation a scalar AR(1) with industry-specific ρ_i governing both persistence and variance would predict.

Oscillatory entry dynamics across industries (BDS, 1978–2022) For each of 19 NAICS sectors, the linearly detrended net entry rate exhibits two features outside the AR(1) class: four sectors show significantly negative lag-4 autocorrelation outside the 95% Bartlett (1946) band (incompatible with any scalar AR(1)), and Fisher's g-test rejects no deterministic periodicity at 5% in 10 of 19 sectors with peaks near $T \approx 8$–9 years. The sign-changing autocorrelation pattern is also harder to reconcile with vintage-capital replacement, which produces positive entry waves rather than alternation. Online Appendix D.2 provides the full sector-by-sector results.

Post-shock periodicity in historical impulse responses BEA investment-in-structures data (1930–2024) tests whether one-time shocks trigger oscillations at industry-specific structural frequencies. Manufacturing investment after the Great Depression/WWII oscillates at $T = 5.0$ years (Figure 5; Fisher's g-test $p = 0.036$ in 1934–1973; $p = 0.578$ in 1974–2013), while mining/oil investment after the 1979 crisis oscillates at $T = 3.25$ years (Figure 6; $p = 0.016$); detected periods are invariant to $\pm$3-year window shifts. The cross-industry frequency difference is consistent with different (φ, μ) across industries, with the vintage-capital alternative (Kydland and Prescott, 1982; Cooper et al., 1999) compatible with rather than competing with the timing-search interpretation (Online Appendix D.3).

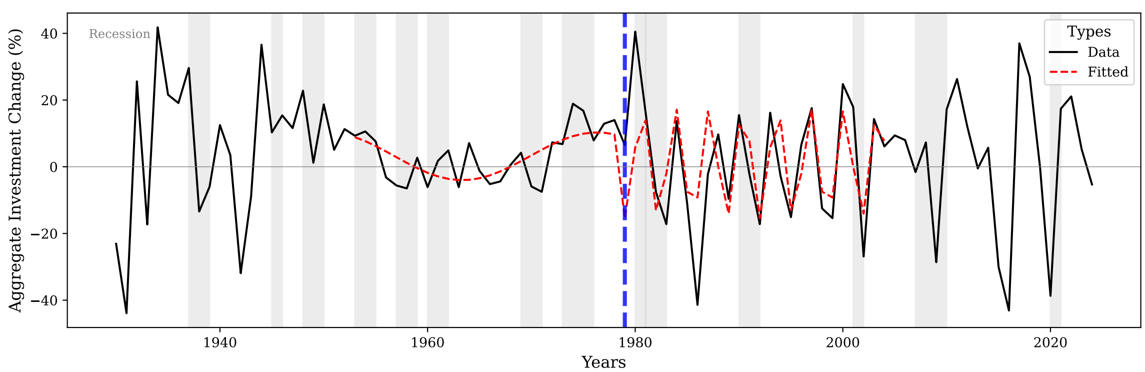
The post-COVID natural experiment Timing search predicts displaced industries overshoot the pre-shock level rather than converge monotonically: the ironing condition forces the economy past optimal scale before cycling back. After the 2020 shock, 14 of 19 BDS NAICS sectors (74%) overshoot above their pre-COVID linear trend in 2021 or 2022, against an AR(1)-monotone-convergence baseline of approximately 30% (binomial test $p < 0.001$). The test is qualitative; the structural

Figure 5: Post-shock periodicity: manufacturing investment



Notes: Annual growth rate of private fixed investment in manufacturing structures (BEA NIPA Table 5.4.1, line 16). Thick solid line: data. Red dashed line: fitted cosine from Fisher's g-test. Gray bands: NBER recessions. Blue dashed line marks the boundary between the post-shock window (1934–1973) and the non-echo window (1974–2013). The fitted cosine tracks the data closely in the post-shock window ($T = 5.0$ years, $p = 0.036$), capturing the regular boom-bust pattern following the Great Depression and WWII mobilization.

Figure 6: Post-shock periodicity: oil and mining investment



Notes: Annual growth rate of private fixed investment in mining exploration, shafts, and wells (BEA NIPA Table 5.4.1, line 24). Blue dashed line marks the 1979 oil crisis. Post-crisis oscillations at $T \approx 3.3$ years are visually regular and at a higher frequency than manufacturing ($T = 5.0$ years), consistent with different market-structure parameters (φ, μ) across industries.

prediction that overshoot magnitude correlates with concentration is not borne out, with sector-specific COVID exposure dominating cross-sectional variation in response magnitude.

Plucking pattern: recovery speed scales with recession depth Isochrony plus amplitude-proportional recovery velocity (Section 3.5) implies recovery speed scales with depth while recovery time does not. For each of seven NBER recessions in FRED INDPRO 1972–2024, OLS regression of average monthly log growth (first 12 months post-trough) on peak-to-trough log decline gives slope $= +0.046$ ($p = 0.029$, $R^2 = 0.65$, Pearson $r = +0.80$). Deeper recessions are followed by faster recoveries, the timing-search analogue of the plucking pattern documented in Friedman (1993); Bordo and Haubrich (2017) and Dupraz et al. (2019).

6 Welfare and Industrial Policy

This section evaluates the welfare cost of timing-search cycles in closed form (Section 6.1) and traces the policy implications of the frequency-welfare separation, including a CHIPS-Act counterfactual (Section 6.2).

6.1 Welfare cost of endogenous cycles

Under the linear-utility household specification of Section 2, household consumption equals aggregate output: $C(t) = n(t)\,\pi(n(t))\,\xi(\dot{n}(t))$. A per-firm time-averaged output deviation from steady state therefore translates directly into a household consumption loss per active firm; with discount rate ρ, steady-state population n_p, and steady-state per-firm output π_0, the household welfare loss from cycling is $n_p\,\pi_0\,\rho^{-1}\,\mathbb{E}[\mathcal{L}]$ to leading order (since $\mathcal{L}$ is the proportional output deviation). We use "per-firm welfare cost" throughout this section as shorthand for the per-firm output deviation $\mathcal{L}$, which is welfare-equivalent under linear utility.

The ironing condition provides a natural welfare measure. Along any equilibrium orbit, the equilibrium value level $E = \frac{\varphi}{2}A^2$ is constant by the ironing condition (Equation (11)), and the time-averaged effective output is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n(t))\,\xi(\dot{n}(t))\,\mathrm{d}t = C = \pi_0\,e^{-E} = \pi_0\,\exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{20}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the time-averaged per-firm effective output is strictly below the optimal-scale, zero-adjustment level. The proportional per-firm effective-output welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2}\,A^2 \quad \text{for small } A. \tag{21}$$

The welfare cost depends on two structural parameters: φ (the curvature of the profit landscape, determining how costly deviations from optimal scale are) and A (the cycle amplitude, selected in stochastic equilibrium by the shock variance σ_ε^2, the bottleneck friction χ, and the profit-landscape curvature φ). Crucially, $\mathcal{L}$ does not depend on μ. The timing friction determines the *frequency* of cycles but not their welfare cost. Two industries with the same φ and A but different μ cycle at different frequencies yet suffer identical welfare losses.

This gives the Lucas (1987) question a structural answer: where Lucas takes the variance of consumption as a reduced-form object, here $\mathbb{E}[\mathcal{L}] \approx \sigma_\varepsilon^2/(2\chi)$ is a function of measurable primitives (curvature φ from profit landscapes, envelope A from entry-exit cycles), with the welfare cost independent of cycle frequency in the linear small-amplitude limit (Theorem 2; Online Appendix E.1 derives the welfare metric and isolates the supplier-side resource flow $\frac{1}{2}\chi\dot{n}^2$ as a separate accounting object).

Numerical welfare cost per industry Calibrating χ_i for each manufacturing industry from $\sigma_{x,i} = \sigma_{\varepsilon,i}/\sqrt{2\chi_i\varphi_i}$ and computing $\mathbb{E}[\mathcal{L}_i] = \sigma_{\varepsilon,i}^2/(2\chi_i)$ as a fraction of $\pi_{0,i}$ yields a mean welfare cost of 0.39% of π_0 across 10 cached manufacturing industries, with range $[0.12\%, 0.70\%]$—comparable to Lucas (1987)- and Reis (2009)-style consumption-equivalent business-cycle welfare costs (typically 0.1 to 1%). Online Appendix E.2 reports the per-industry table.

6.2 Industrial policy and the anatomy of recovery

The frequency-welfare separation (Theorem 2) delivers an instrument-to-outcome map under the per-firm effective-output welfare metric. The structural parameters act on identifiable margins: μ governs timing alone, φ governs timing and severity, χ governs severity and welfare, σ_ε governs severity and welfare, and κ governs only the steady-state ironing level. In the stochastic equilibrium the per-firm welfare cost reduces to $\mathbb{E}[\mathcal{L}] \approx (\varphi/2)\mathbb{E}[A^2] = \sigma_\varepsilon^2/(2\chi)$: both φ and μ cancel, and κ is absent. Table 3 summarizes the taxonomy.

Welfare-irrelevant policy The taxonomy makes a substantive contribution to policy discourse: *many seemingly stabilization-relevant policies do not move welfare in*

Table 3: Policy anatomy: instrument-to-outcome map

Lever	Frequency	Severity	Welfare	Example
μ (matching friction)	✓	—	—	Permitting reform
φ (profit curvature)	✓	✓	—	Cluster subsidies
χ (pipeline congestion)	—	✓	✓	Pipeline capacity
σ_ε (shocks)	—	✓	✓	Demand stabilization
κ (sunk cost)	—	—	—	Per-firm transfer

Notes: ✓ marks a primitive that moves the column outcome under the per-firm effective-output welfare metric of Section 6.1; — marks no effect. The frequency $\omega = \sqrt{\varphi/\mu}$, the stationary severity $\sigma_x = \sigma_\varepsilon/\sqrt{2\chi\varphi}$, and the per-firm welfare cost $\mathbb{E}[\mathcal{L}] = \sigma_\varepsilon^2/(2\chi)$ are the closed-form expressions from Theorem 2. Welfare-cost effects are stated under the maintained per-firm metric; alternative metrics that include the supplier's resource flow leak μ into the welfare formula at order 10–25% (Online Appendix E.1).

this framework. Faster permitting, e-permitting reform, and entry-matching infrastructure investment all move μ, accelerating cycles but leaving the per-firm welfare cost unchanged. Cluster subsidies and agglomeration support move φ, accelerating cycles *and* lowering severity σ_x, yet the severity reduction is exactly offset by the curvature increase: $\mathbb{E}[\mathcal{L}] = (\varphi/2)\mathbb{E}[A^2] = (\varphi/2) \cdot \sigma_\varepsilon^2/(\chi\varphi) = \sigma_\varepsilon^2/(2\chi)$, independent of φ. The Lucas (1987) intuition—that lowering aggregate volatility raises welfare—survives in attenuated form: only the volatility channel that loads onto $(\sigma_\varepsilon, \chi)$ moves welfare cost, while the volatility channel that loads onto φ does not. A policymaker who interprets faster cycles or lower amplitude as a welfare improvement may be reading off the wrong margin.

CHIPS Act counterfactual Calibrating $(\varphi, \mu, \chi, \sigma_\varepsilon)$ for NAICS 3344 (Semiconductor and Other Electronic Component Manufacturing) from BDS 1978–2022 yields a baseline spectral period $T = 4.75$ years matching the observed BDS net-entry-rate peak. Table 4 reports the model-implied effects of three stylized interventions and one combination, all relative to baseline.

The model contains two structurally distinct entry-cost objects: κ, the per-firm sunk cost paid at entry, and χ, the bottleneck friction summarizing bottleneck-strain dissipation. The welfare formula depends on χ alone; κ enters the ironing level but not the welfare loss. Whether a real-world entry-cost intervention loads onto χ or κ is an institutional question—a fixed entry fee or per-firm transfer is pure κ; a subsidy that pays for construction and permitting plausibly moves χ via the convex

Table 4: Counterfactual policy interventions, NAICS 3344

Scenario	Period ratio	Amplitude ratio	Welfare ratio
A. CHIPS-style subsidy: $\chi \times 0.5$	1.00	1.41	2.00
B. Cluster policy: $\varphi \times 1.2$	0.91	0.91	1.00
C. Streamlined permitting: $\mu \times 0.7$	0.84	1.00	1.00
A + B combined	0.91	1.29	2.00

Notes: Calibration and full taxonomy in Online Appendix E.3. Scenario A halves the bottleneck friction χ, a stand-in for an entry subsidy that subsidizes construction and permitting and so weakens the bottleneck-strain mechanism: cycle period unchanged, amplitude rises by $\sqrt{2}$, welfare cost doubles. Scenario B raises φ by 20%: cycle shortens by 9%, amplitude falls by the same factor, welfare unchanged. Scenario C reduces μ by 30%: cycle shortens by 16%, amplitude and welfare unchanged. Combined A+B speeds the cycle and raises both amplitude and welfare cost.

bottleneck-strain mechanism. The qualitative claim that some real-world entry friction is stabilizing is conditional on the friction loading onto χ rather than κ.

The taxonomy is sharper than what existing endogenous-cycle frameworks deliver, where frequency and welfare cost are typically governed by the same primitives and any frequency-altering policy moves welfare. Timing search separates these channels, letting policymakers ask "which policy speeds up cycles?" and "which policy stabilizes welfare?" as separable questions with different answers.

7 Discussion

The closed-form frequency $\omega = \sqrt{\varphi/\mu}$ and its separation from the welfare cost of cycles rest on five maintained assumptions: a hump-shaped profit landscape, the Gaussian specification, linear utility, a convex matching friction, and exogenous exit. This section addresses each in turn, showing that the cycle mechanism survives relaxations while the closed-form exactness is Gaussian-specific, and outlines extensions.

7.1 Modeling choices and limitations

Four limitations bound the scope of the closed-form result.

(i) Gaussian exactness. The symmetric-cycle prediction of the Gaussian specification is counterfactual; under general hump-shaped profits (Online Appendix B.2), cycles become asymmetric and the frequency formula is accurate to 1.6% at empirically relevant amplitudes (Online Appendix B.4 and C.2).

(ii) Exogenous exit. The model treats exit through the constant-rate δ rather than as a learning- or productivity-driven endogenous decision; combining timing search with Jovanovic (1982)-style learning is a natural extension.

(iii) Linear utility and the per-firm welfare metric. Under CRRA preferences or under welfare metrics that include the supplier's resource flow, (φ, μ) leak into the welfare cost with median magnitude 10–25% (Online Appendix E.1), so the frequency-welfare separation is exact under our maintained metric and approximately exact otherwise.

(iv) Within-manufacturing scope of the empirical identification. The headline test isolates the φ channel under the maintained restriction that μ is approximately uniform across manufacturing 3-digit industries; the substantive treatment is in Section 5.3, with three lines of evidence consistent with the restriction (the recovered $\hat{\beta}_\gamma \approx 0.45$ aligning with the structural $1/2$, external μ proxies returning wrong-signed coefficients, and the microfounded $\mu_i = (1-\alpha)/[\alpha(\delta_i n_{p,i})^2]$ predicting narrow within-manufacturing variation). A direct test of the μ channel outside manufacturing requires industry-friction-specific proxies at NAICS-3 resolution that current public data do not provide and is left for future data work.

7.2 Extensions

Several extensions merit investigation: combining timing search with Jovanovic (1982)-style learning to endogenize exit (producing entry waves followed by shakeouts); relaxing linear utility to let industries interact through aggregate consumption; endogenizing (φ, μ) through industry investment in linkages or entry infrastructure; and combining timing search with financial frictions (Beaudry et al., 2020) to produce models with multiple spectral peaks (one structural, one from financial amplification).

8 Concluding Remarks

This paper provides a structural account of cross-industry frequency heterogeneity. Under *timing search*—agents choosing *when* to enter a market subject to matching frictions—the equilibrium ironing condition forces overshoot rather than monotone adjustment, and the closed-form cycle frequency $\omega_0 = \sqrt{\varphi/\mu}$ emerges in two market-structure primitives: profit-landscape curvature φ and entry-matching friction μ.

The headline empirical test brings this prediction to U.S. industry data: across 16 manufacturing 3-digit industries, the cross-industry slope $\hat{\beta}_\gamma$ falls in the $[0.41, 0.48]$ range across estimators (multitaper, Burg AR(12), and IV with historical concentration from Kim (1995) instrumenting for current γ), broadly consistent with the structural prediction $\beta_\gamma = 1/2$. Given a moderate first stage ($F = 5.30$ with the 1947 Hoover instrument), we read the IV as corroborative rather than independently identifying; under Anderson-Rubin weak-IV-robust inference, 1/2 is not rejected ($p = 0.78$) and 0 is rejected at $p = 0.007$. Multi-sector RBC (Acemoglu et al., 2012) and Khan and Thomas (2008)-style heterogeneous capital-adjustment costs reproduce cross-industry frequency dispersion but not the empirical slope across estimators. Cycle length, in this account, is a property of how firms interact within an industry, not of the shocks hitting it.

A second result is a *frequency-welfare separation*: $\omega_0 = \sqrt{\varphi/\mu}$ depends only on the market-structure primitives $\{\varphi, \mu\}$, while the per-firm effective-output welfare cost $\mathcal{L} = \sigma_\varepsilon^2/(2\chi)$ depends only on the disjoint pair $\{\sigma_\varepsilon, \chi\}$. The separation emerges from the same ironing condition rather than being imposed, and it delivers an instrument-to-outcome map for industrial policy: μ governs timing alone, φ governs both timing and severity, χ governs severity and per-firm welfare, and κ governs only the ironing level (Section 6.2).

More broadly, timing search recasts where cycle frequency comes from: industries differ in their characteristic frequencies because they differ in market structure, not because they are hit by different shocks. The same logic—agents choosing *when* to act under matching frictions—extends naturally to capital investment, hiring, technology adoption, and financial-market participation, each with its own market-structure-pinned frequency, and the closed form $\omega = \sqrt{\varphi/\mu}$ provides a template: a structural cycle object measurable from primitives, with welfare and policy implications that follow from the same disjoint determinants that pin frequency itself.

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